universidade de aveiro | deti departamento de electrónica, telecomunicações e informática

42088 – Industrial Project

Project Status Report

Milestone 2 – Elaboration

Project name:	Automatic, universal push-button actuator, equipped with force gauge
Customer:	Altice Labs
Customer supervisor:	Eng. Hélder Alves Eng. Nuno Balseiro
Date issued:	06/11/2024
Team members:	Coordinator: Cláudia Almeida, claudiasalmeida@ua.pt, 914102617. Other team members: Rafael Alves, rafael01@ua.pt , 937353627. Gonçalo Peralta, goncaloperalta@ua.pt , 916828627. Pedro Matias, pedrodsmatias@ua.pt , 917126548.
Supervisor:	Professor Pedro Fonseca
Class supervisor:	Professor Pedro Fonseca

Management summary:

The project is on track to meet its delivery deadlines. The team has successfully finished the Inception phase and is now actively working on the Elaboration phase. The component list, hardware circuit designs, and use case definitions are all now being finalized. There have been no substantial delays, and all the important components are being scheduled for purchasing within the current timeframe. As a consequence of the most recent meetings with the customer and the project supervisor, where the more technical issues were discussed, the electronic components were picked to ensure the product's flawless operation, resulting in an adjustment to the final budget. As we move into the Construction phase, the integration of the actuator, force sensor, and API, with iterative testing will be closely monitored, intended to address any compatibility issues early on.

Architecture

The architecture of the system consists of three main components: the **Control Unit**, the **Control Electric Circuit**, and the **Mechanical Actuation**.

- The **Control unit** is managed by a Raspberry Pi that runs the control software and serves as the central processing component for the system. The Raspberry Pi generates a PWM (Pulse

Width Modulation) signal to regulate the actuator's movement, allowing for controlled button pressing with variable force. It also maintains a Secure Shell (SSH) connection with the DUT, allowing it to detect feedback from the device in real time, such as system responses or status changes upon a button click. For force measurement, the Raspberry Pi reads data from the force sensor using I2C/SPI protocols, ensuring accurate and timely data transfer from the sensor to the control software.

- The **Control electric circuit** amplifies the PWM signal generated by the Raspberry Pi, ensuring sufficient power is supplied to the actuator. This circuit includes an **H-Bridge**, allowing current to flow in both directions as required to control the actuator's movement accurately. The circuit, powered by a 12V, 3A DC power source, will provide the energy needed to drive the actuator with stability and precision, allowing for both pressing and releasing movements, required for testing various types of buttons.
- The **Mechanical Actuation** assembly is designed to interface directly with the DUT, positioning the actuator over the buttons to be tested. This component includes a clamp that securely attaches to the DUT, providing flexibility and adaptability for different button placements and device sizes. Mounted within the clamp is a linear actuator that performs the physical pressing of the buttons, with a force sensor placed behind the actuator to capture the exact force applied during each press. This assembly connects to a PCB board, which relays signals from the force sensor back to the Control Unit, ensuring accurate control and data capture throughout the testing process.

Figure 1 shows the improved and updated mechanical design of the product.

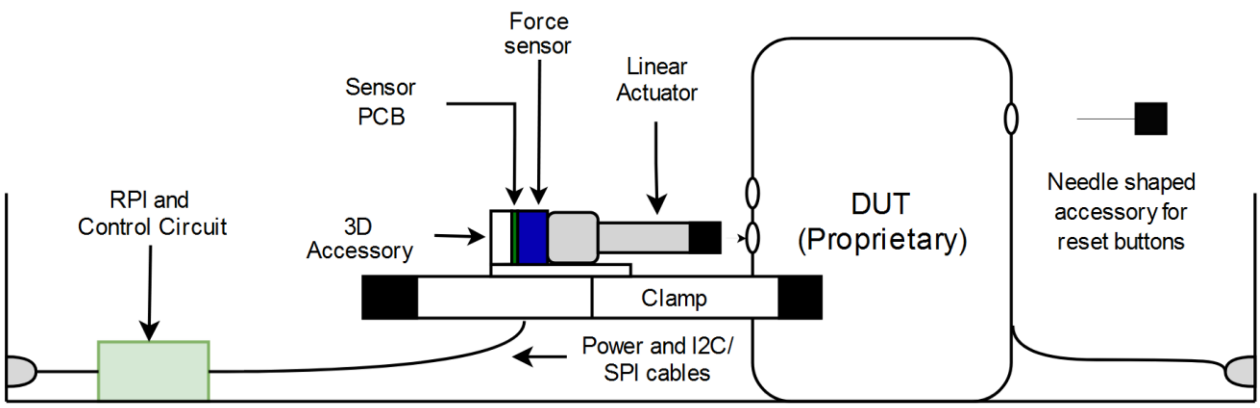

Figure 1. Mechanical assembly of the product.

Figure 2 shows the Unified Modelling Language (UML) Sequence Diagram, which represents the interaction amongst the different components of the system.

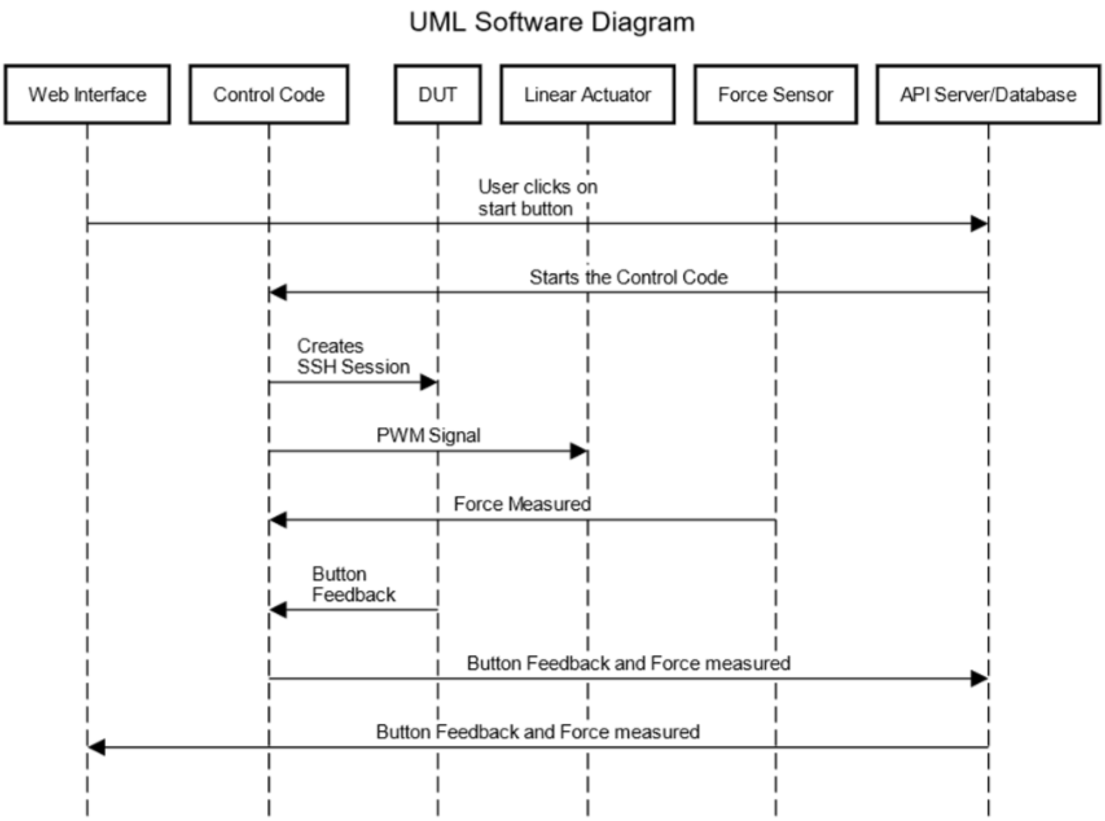

Figure 2. UML Diagram.

Key milestones

ID	Description	Original planned completion date	Current planned completion date	Actual completion date
Project Vision	Market and needs to which the project responds	23/10/2024	---	23/10/2024
Project Plan	Working Phases, Budget and Risk List	23/10/2024	---	23/10/2024
Mock Up	A physical model of the system	23/10/2024	---	23/10/2024
Hardware	Final Component Lists	30/10/2024	5/11/2024	6/11/2024
Project Status Report – M2 Elaboration	A summary of the current project status	6/11/2024	6/11/2024	6/11/2024

Project Use Cases – M2 Elaboration	Description of user interaction with the product	6/11/2024	6/11/2024	6/11/2024
Software Web Interface	Suggested visual presentation of the web interface	13/11/2024	13/11/2024	
Hardware PCB's design	Design of the system's printed circuits and electrical circuits	13/11/2024	20/11/2024	

Progress and deviation from plan

The project has generally progressed well; however, finalizing and structuring the component list took an additional seven days beyond the original timeline. This delay has impacted the timeline for requesting and purchasing the components, which may cause slight setbacks in the hardware assembly phase. To overcome this impediment of progress on the hardware testing side, we're going to move forward with the printed circuit boards, which are a fundamental part of starting tests, and start working and simulating data reception on the Raspberry Pi's web interface.

Work plan for next iteration

In the next iteration, our primary focus will be on the **design of the hardware printed circuit boards (PCBs)** to support the control electric circuit for the actuator and force sensor. As the team anticipates some waiting time for component deliveries, we plan to adjust our schedule by beginning **web page development for the Raspberry Pi** ahead of schedule and design some visual presentation of the Web interface itself in various stage of its use. This will allow us to lay the foundation for the control interface, enabling future integration with the actuator and force sensor. Table 2 shows the sprint plan updated.

Sprints	Weeks	Tasks	Claúdia	Rafael	Pedro	Gonçalo
1	3	Detailed system architecture and list of components finalized	x	x	x	x
		Hardware printed board design		x	x	
		Web page development on the Raspberry Pi	x			x
2	3	Test the load cell/resistive sensor to measure the drive force	x			x
		Integrating the actuator and sensor into the mechanical design Testing basic actuator control using the Raspberry Pi	x	x	x	x
		Test the API's interaction with the actuator and force feedback. Test universal adjustment on various test devices (routers/gateways)	x	x	x	x
3	3	Ensure accurate force measurement with load cells			x	x
		Testing the API, actuator feedback and force gauge functionality. Debug final problems related to the software or the mechanical fit. Fully functional prototype	x	x	x	x
4	3	Collecting final feedback before final construction	x	x	x	x
		Delivery of the prototype to the client	x	x	x	x

Table 2. Work sprints plan and schedule.

Financial status

Following the development of the system's architecture in the previous iteration report, the team undertook a more in-depth investigation of the components needed to achieve the project specifications. This extensive analysis proved that our initial component selections were generally correct. However, it resulted in a significant rise in the overall budget.

One of the main factors contributing to this rise is the necessity for components with operating ranges and characteristics that better match our project's requirements, which are much more expensive. For instance, we wanted to replace the linear actuator to a type with a lower maximum force to ensure compatibility with our system, but this model costs roughly triple the amount. Similarly, we chose a more precise force sensor because force measurement is crucial for our product. This sensor, while more expensive, will significantly improve the accuracy and reliability of test results.

The remaining increase in the overall cost is due to small and inexpensive components that we did not include in the previous presented budget.

Quantity	Description		Status	Price Per Unite(€)	Total Value(€)	% of Total Value
1	Honeywell FMAMSDXX015WC2C3	Force Sensor	Planned	33,67	33,67	18%
1	Actuonix L12-50-50-12-S	Actuator	Planned	83,66	83,66	45%
2	JST Comercial RE-H022TD-1190	Actuator connector to PCB	Planned	0,18	0,36	0%
1	CUI Inc. SMM30-12-K-P6	Power Supplier	Planned	17,76	17,76	10%
1	Same Sky PJ-079BH	Connector to Power Supplier	Planned	1,5	1,5	1%
1	Raspberry Pi 4 B	Processing Unit	Planned	32,55	32,55	17%
1	ST L6205D	Controler for the actuator	Planned	6,12	6,12	3%
1	Nichicon GYF1H101MCQ1GS	Capacitor	Planned	1,11	1,11	1%
1	KYOCERA AVX KAM21BR71H104JT	Capacitor	Planned	0,19	0,19	0%
1	TAIYO YUDEN MAASU21GSB7224KTCA01	Bootstrap Capacitor	Planned	0,4	0,4	0%
1	KYOCERA AVX KGF15AR71H103KT	Bootstrap Capacitor	Planned	0,23	0,23	0%
2	KYOCERA AVX 12065A562FAT2A	Capacitor	Planned	0,94	1,88	1%
2	onsemi / Fairchild 1N4148	Bootstrap Diode	Planned	0,09	0,18	0%
2	Panasonic ERA-8VRB1003V	Resistor	Planned	1,18	2,36	1%
1	Vishay CRCW2010100RJNEFIF	Bootstrap Resistor	Planned	0,55	0,55	0%
1	TI INA381A3	Current Amplifier	Planned	1,1	1,1	1%
1	TE Connectivity RLC73K2ER162FTDF	Current Measuring Resistor	Planned	0,46	0,46	0%
1	Vishay MCS0402MD1002BE100	Pull-Up Resistor	Planned	0,45	0,45	0%
1	CTS Electronic Components 14TR08SA103CP	Potentiometer	Planned	0,71	0,71	0%
1	TI TLV9032QDGKRQ1	Capacitor	Planned	1,5	1,5	1%
1	KYOCERA AVX KAM21BR71H104JT	Capacitor	Planned	0,19	0,19	0%
				Total:	186,93	100%

Group contribution

Group member	Major contributions	Work share (%)
Cláudia Almeida	Team coordinator, responsible for organizing regrouping sessions and managing communication with external parties. Primary editor of written documents, developed the use cases, studied market relevance for industrial automation, and led Milestone 1 presentation.	25
Rafael Alves	Developed the final hardware components list, analysed project risks for each phase and proposed mitigation strategies, and led Milestone 1 presentation.	25
Gonçalo Peralta	Created all system diagrams and drafts of the mechanical product for documentation, developed the block diagrams and use cases, sprints spread calendar and led the initial planning for the software approach to the project.	25
Pedro Matias	Contributed to researching and finalizing the hardware components list and supported the assessment of technical needs for project phases.	25

Comments and remarks

Attached are the documents relating to the risk analysis of the project and the block diagrams of the system broken down into various levels.